

HW 2 Solutions to non-graded problems

Problem 5

b

The joint likelihood function for a sample $X_1, \dots, X_n$ is $f(x_1, \dots, x_n; \theta) = \theta^n (x_1 \dots x_n)^{\theta-1}$, with log-likelihood function $n \log \theta + (\theta - 1) \log(x_1 \dots x_n)$. Differentiating with respect to θ and equating the derivative to 0, we get

$$n/\theta + \log(x_1 \dots x_n) = 0,$$

so the MLE for θ is

$$\hat{\theta} = \frac{-n}{\log(\prod_{i=1}^n X_i)}.$$

c

As explained in the section, the MME estimator for θ is $\tilde{\theta} = \bar{X}/(1 - \bar{X})$. Therefore, the estimators for these data samples are $\hat{\theta} = 0.5492604, 2.210124, 0.9588026$ and $\tilde{\theta} = 0.5974696, 2.400435, 0.8645585$ (in order).

Problem 6

b

We first calculate the expected value of X . Since the pdf is symmetric about $\theta/2$, $\theta/2$ must be the mean. We can also calculate it directly:

$$\begin{aligned} \mathbb{E}X &= \int_0^{\theta/2} (4/\theta^2)x^2 dx + \int_{\theta/2}^{\theta} -(4/\theta^2)x^2 + (4/\theta)x dx \\ &= \theta/6 + \theta/3 \\ &= \theta/2. \end{aligned}$$

Now we equate the expected value of X with the first moment of the sample, $\bar{X}$: $\theta/2 = \bar{X}$. Solving for θ gives the MME estimator

$$\tilde{\theta} = 2\bar{X}.$$

c

$$\tilde{\theta} = 0.74646.$$